

Total No. of Questions : 10]

SEAT No. :

P2046

[Total No. of Pages : 2

[5059] - 651

B.E. (Computer Engineering)

MULTIDISCIPLINARY NATURAL LANGUAGE PROCESSING

(2012 Pattern) (Elective - II) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:-

- 1) Answer Q1 or Q2 , Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data if necessary.

Q1) a) Define the following terms: **[6]**

i) Parsing ii) Robust Parsing

b) Explain Automatic Stochastic Tagging of natural language texts. **[4]**

OR

Q2) a) Explain the significance of Lexical Knowledge Networks in NLP. **[6]**

b) Write a short note on : Viterbi algorithm of FSM process. **[4]**

Q3) a) Illustrate the Basic steps of EM algorithm alongwith it's Analysis. **[6]**

b) Explain in short : Forward Backward probability.. **[4]**

OR

Q4) a) Define HMM. Draw the diagram that shows Ergodic and non-ergodic models **[6]**

b) What is Conditional Random Field (CRF)? How it is advantages than HMM? **[4]**

- Q5)** a) Explain the following levels of Speech chain: [8]
i) Linguistic level
ii) Physiological level
iii) Acoustic level
b) What is Speech Disorders ? Give it's Causes. [8]

OR

- Q6)** a) Explain Sentence-level Phenomena. Explain how it differs from adaptation level phenomenon. [8]
b) Define Speech Perception. Explain it's role in recognition of speech. [8]
- Q7)** a) What do you mean by Multilingualism? Differentiate between Multilingualism and bilingualism, [8]
b) Explain in short the following approaches of Word Sense Disambiguation. [8]
i) Knowledge-Based
ii) Supervised
iii) Unsupervised

OR

- Q8)** a) Explain Multilingual Dictionary. Also Explain automatic detection of Multilingual Dictionaries on the Web. [8]
b) Write a note on: Semantic role [8]
- Q9)** a) Explain the following terms of NLP applications: [10]
i) Web mining
ii) Text mining
b) What is Cross Lingual Information Retrieval ? Explain it. [8]

OR

- Q10)** Write a Note on: [18]
a) Robust and Scalable Machine Translation
b) Question Answering in Multilingual Setting
c) Sentiment Analysis

